

Project #7 Report by Tomas Salaz
November 26, 2024

1) Derivative Approximation

1) $U = U(x)$, use Taylor Series to find formulas for derivatives, order 7

a) $U'(x_i)$ using x_i, x_{i+1}

$*C = x_{i+1} - x_i \leftarrow \text{Spacing}$

$$U(x_{i+1}) = U(x_i) + C \underbrace{U'(x_i)}_{\text{ISOVATE}} + C^2 \frac{U''(x_i)}{2!} + C^3 \frac{U'''(x_i)}{3!} + O(C^4)$$

$$U'(x_i) = \frac{U(x_{i+1}) - U(x_i)}{C} - \frac{C}{2!} U''(x_i) - \frac{C^2}{6} U'''(x_i) + O(C^3)$$

NOTE: See 1st Order Approx $\Rightarrow U'(x) = \frac{f(x+h) - f(x)}{h}$ for h

$$U'(x_i) = \frac{U(x_{i+1}) - U(x_i)}{C}$$

b) $U'(x_i)$ using x_{i-1}, x_i $C = x_i - x_{i-1} \leftarrow \text{Spacing}$

$$U(x_i) = U(x_{i-1}) + C \underbrace{U'(x_{i-1})}_{\text{ISO}} + C^2 \frac{U''(x_{i-1})}{2!} + C^3 \frac{U'''(x_{i-1})}{3!} + O(C^4)$$

$$U'(x_i) = \frac{U(x_i) - U(x_{i-1})}{C} - \frac{C}{2!} U''(x_i) + \frac{C^2}{3!} U'''(x_i) + O(C^3)$$

$\rightarrow$ Truncate

$$U'(x_i) = \frac{U(x_i) - U(x_{i-1})}{C}$$

c) $U'(x_i)$ using x_{i-1}, x_i, x_{i+1}

$$U(x_{i+1}) = U(x_i) + C U'(x_i) + C^2 \frac{U''(x_i)}{2!} + C^3 \frac{U'''(x_i)}{3!} + O(C^4)$$

$$U(x_{i-1}) = U(x_i) - C U'(x_i) + C^2 \frac{U''(x_i)}{2!} - C^3 \frac{U'''(x_i)}{3!} + O(C^4)$$

$$U(x_{i+1}) - U(x_{i-1}) = 2C U'(x_i) + C^3 \frac{U'''(x_i)}{3!} + O(C^5)$$

$$U'(x_i) = \frac{U(x_{i+1}) - U(x_{i-1}))}{2h}$$

d) $U'(x_i)$ using $x_i, x_{i+1}, x_{i+2}, x_{i+3}$

$$U(x_{i+1}) = U(x_i) + C U'(x_i) + C^2 \frac{U''(x_i)}{2!} + C^3 \frac{U'''(x_i)}{3!} + O(C^4)$$

$$U(x_{i+2}) = U(x_i) + 2C U'(x_i) + 2C^2 \frac{U''(x_i)}{2!} + 8C^3 \frac{U'''(x_i)}{3!} + O(C^4)$$

$$U(x_{i+3}) = U(x_i) + 3C U'(x_i) + 9C^2 \frac{U''(x_i)}{2!} + 27C^3 \frac{U'''(x_i)}{3!} + O(C^4)$$

- Solve for $U'(x_i)$ using a linear combination

$$U'(x_i) = \frac{-11U(x_i) + 18U(x_{i+1}) - 9U(x_{i+2}) + 2U(x_{i+3}))}{6h}$$

-The Approximations were done by hand (images to the left) using the method in class. I followed the same procedure but used 'c' instead of 'h' for some reason, so we have big $O(c)$ notations. But I will be referring to this notion in classic $O(h)$ notation.

Findings and Observations

The order of the approximations are as follows:

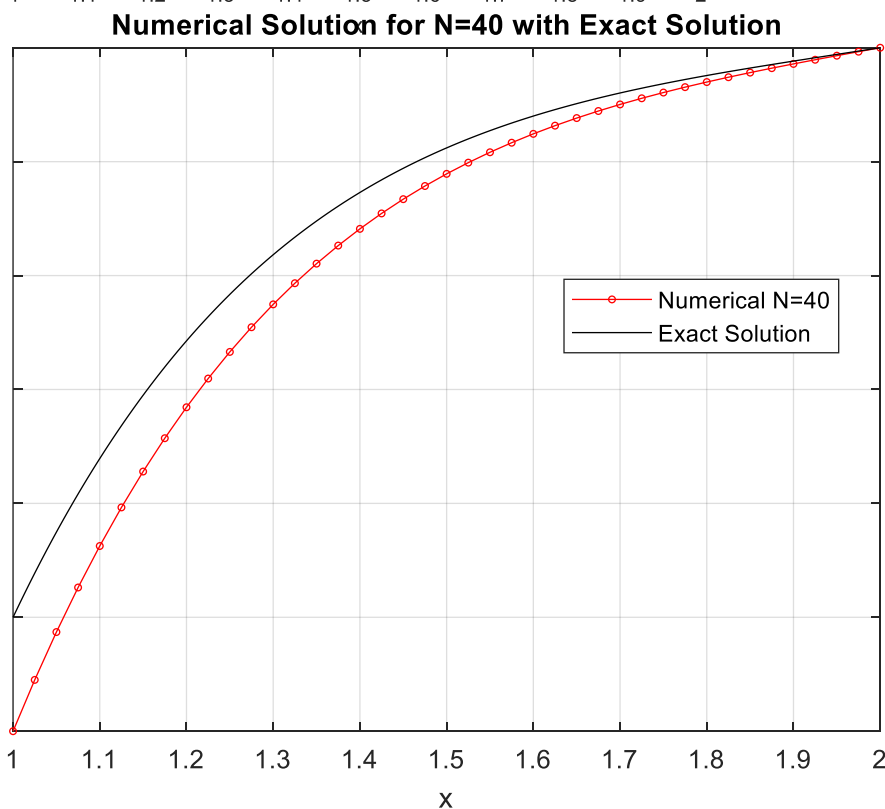
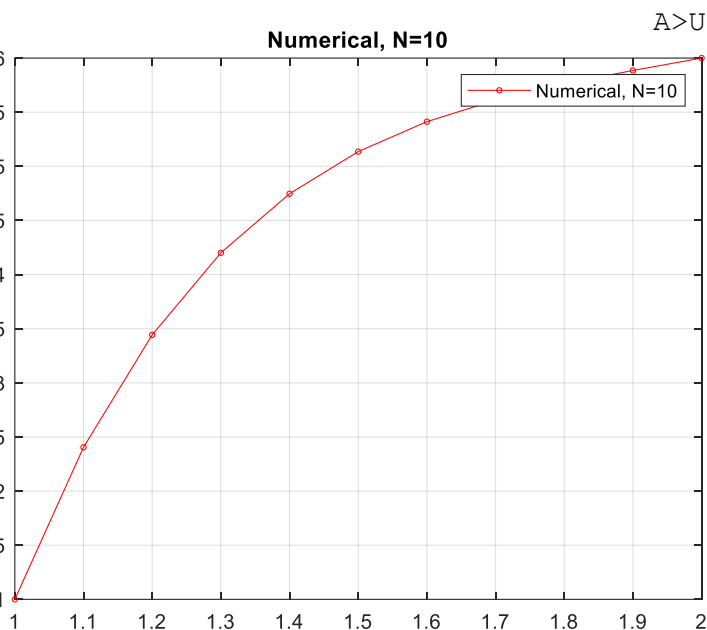
- $x_i, x_{i+1} \rightarrow O(h)$
- $x_{i-1}, x_i \rightarrow O(h)$
- $x_{i-1}, x_{i+1} \rightarrow O(h^2)$
- $x_i, x_{i+1}, x_{i+2}, x_{i+3} \rightarrow O(h^3)$

This shows that the order isn't dependent on the number of x_i 's that are used but is dependent on the distance from x_i . This implies that the error can be influenced by the Spacing 'h' and the symmetry between the points. From this we can infer that more points with closer spacing yields better accuracy.

2) Numerical Solutions

Use (*) or (**) to get an approximation for

$$u''(x) + 3u'(x) + 2u(x) = 4x^2 \quad u(1) = 1 \quad u(2) = 6$$



I also noticed that the Approximation for N=40 contains all the N=10 Nodes, which shows that there is consistency in my Solution Approximation implementation.

3) Numerical Integration and Richardson Extrapolation
 A and B> Determine Number of subintervals, N, to have an error less than 10^{-4} , using that N find the exact error.

3a) Determine Number of subintervals to approx

$\int_0^{\pi} \sin(x) dx$ within 10^{-4} using CTR.

Error Formula $\rightarrow |E| = -\frac{(b-a)^3}{12N^2} \cdot \frac{|f'''(x)|}{6}$ $b-a = \pi$ $|f'''(x)| < k$
 $|f'''(x)| < 1 \rightarrow$ Bounded by $\sin(x)$

$f(x) = \sin(x)$

$f'(x) = \cos(x)$

$f''(x) = -\sin(x)$

$10^{-4} \leq -\frac{(b-a)^3}{12N^2}$

Solve for N $\rightarrow N \geq \sqrt{\frac{\pi^3}{12(10^{-4})}}$

$N \geq 161$

$N \geq 160.74$

3b) Error for CTR

*Note $\int_0^{\pi} \sin(x) dx = 2$
 Exact Value

Using Trapezoid

$\int_0^{\pi} f(x) dx \approx T(f, P) = h \left[\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2} [f(x_0) + f(x_n)] \right]$

$= \frac{\pi}{161} \left[\frac{0+2}{2} + \sum_{i=1}^{160} f(x_i) \right]$

$= 1.999936540083114$ *USING PYTHON*

Actual Error $\Rightarrow$

$E = |T_n - 2|$

$= |1.999936540083114 - 2|$

$= .000063459916886$ Actual Error

In class we hinted towards the solution for this problem. It was simple, we just had to use the error formula and the given interval to solve for N. Once you rearrange and plug into a calculator, you get 160. 'Some Numbers' so we round up to get 161 subintervals.

To find the exact value was also simple, just use the Composite Trapezoid Rule with 161 Partitions to get a value, then subtract that from the exact values which is trivially known to be 2 to get the exact error.

I used a Python Script to find the CTR value, then double checked in MatLab (included in the coding files for this assignment), the calculated to find the exact error. This error was within the expected range of 10^{-4}

C> Starting from the given equation, increase the accuracy by applying the Richardson Extrapolation 1 time and include the error term.

$$\begin{aligned} \text{Ex) } f'(x) &= \frac{f(x+h) - f(x)}{h} \\ \phi(h) &= \frac{f(x+h) - f(x)}{h} \\ \text{Taylor Expansion of } f(x+h) &= f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + O(h^4) \\ \text{Expand } \phi(h) &\rightarrow \phi(h) = f'(x) + \frac{h}{2} f''(x) + \frac{h^2}{3!} f'''(x) + O(h^3) \\ \text{Expand } \phi(h/2) &= \frac{f(x+h/2) - f(x)}{h/2} = f'(x) + \frac{h}{4} f''(x) + \frac{h^2}{24} f'''(x) + O(h^3) \\ \text{Sub into Richardson Extrap} & \\ \phi_{\text{new}}(h) &= \frac{4\phi(h/2) - \phi(h)}{3} \\ &= \frac{4(f'(x) + \frac{h}{4} f''(x) + \frac{h^2}{24} f'''(x) + O(h^3)) - (f'(x) + \frac{h}{2} f''(x) + \frac{h^2}{3!} f'''(x) + O(h^3))}{3} \\ &= f'(x) + O(h^4) \\ &= f'(x) + O(h^4), \quad * f'(x) = \frac{1}{2} [\phi_{\text{new}}(h)] \\ \phi(h) &= \frac{1}{2h} [8f(x + \frac{h}{2}) - f(x+h) - 7f(x)] + O(h^4) \end{aligned}$$

We started this problem in class and outlined the process for Richardson Extrapolation. We just had to expand the Phi into their Taylor Series Representations, so expanding Phi(h) and Phi(h/2). Then we sub those expansions back into the original Phi(h) function to get a representation. We take a Derivative and then we are done. The process is tedious and requires attention to details but wasn't too complex. Wikipedia says Richardson Extrapolation is a 'Sequence Acceleration' method so repeating this process would lead to a sequence of values.